

## Lecture 5

### The Beta–Bernoulli Conjugate Model

Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta), \quad \theta \sim U(0, 1).$$

Then

$$p(\theta \mid x_1, \dots, x_n) \sim \text{Beta}\left(\sum X_i + 1, n - \sum X_i + 1\right).$$

$$p(\theta \mid x_1 \dots x_n) \propto \theta^{\sum X_i} (1 - \theta)^{n - \sum X_i}$$

$$\text{Beta}(a, b) \propto \theta^{a-1} (1 - \theta)^{b-1}$$

**Conjugate prior.** If the prior distribution is of the same family as the posterior distribution, then we call such a prior a **conjugate prior**.

$$p(\theta \mid X_1 \dots X_n) \propto p(\theta) p(X_1 \dots X_n \mid \theta)$$

$$p(\theta \mid X_1 \dots X_n) \propto \theta^{\alpha + \sum X_i - 1} (1 - \theta)^{\beta + (n - \sum X_i) - 1}$$

$$\therefore \text{Beta}\left(\sum X_i + \alpha, n - \sum X_i + \beta\right)$$

$$p(\tilde{y} = 1 \mid X_1 \dots X_n) = E(\theta \mid X_1 \dots X_n) = \frac{\alpha + \sum X_i}{n + \alpha + \beta}$$

$$\text{As } n \rightarrow \infty, \quad \approx \frac{\sum X_i}{n}$$

**General setup.**  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$ . If  $p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$  [Prior], then  $\theta \sim \text{Beta}(\alpha, \beta)$ .

$$\boxed{\frac{\sum X_i}{n}} \quad \text{Var}(\theta \mid X_1 \dots X_n) = \frac{(\alpha + \sum X_i)(\beta + n - \sum X_i)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}$$

$$\text{As } n \rightarrow \infty, \quad \text{Var}(\theta \mid X_1 \dots X_n) \approx \frac{(\sum X_i)(n - \sum X_i)}{n^2(n)} \rightarrow 0$$

### Normal Distribution with Known Variance

$$p(y \mid \theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$

$$p(\theta \mid y) \propto \underline{\underline{p(\theta)}} \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$

$$\exp(A\theta^2 + B\theta + C) \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$

$p(\theta) \sim N(\mu_0, \tau_0^2)$ , & then,

$$p(\theta \mid y) \propto \exp\left(-\frac{1}{2\tau_0^2}(\theta - \mu_0)^2\right) \exp\left(-\frac{1}{2\sigma^2}(y - \theta)^2\right)$$

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where}$$

$$\mu_1 = \frac{\frac{1}{\tau_0^2} \mu_0 + \frac{1}{\sigma^2} y}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \quad \text{and} \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

**Show:**

Now let  $y_1, \dots, y_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$ .

$$\bar{y} \sim N\left(\theta, \frac{\sigma^2}{n}\right)$$

$$p(\theta \mid \bar{y}) \propto p(\theta) p(\bar{y} \mid \theta)$$

$$\therefore \theta \mid y_1, \dots, y_n \text{ (equivalently } \theta \mid \bar{y})$$

$$\therefore \theta \mid \bar{y} \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n}{\sigma^2} \bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \quad \& \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

$$p(y \mid \theta) \propto \exp\left\{-\frac{\sum (y_i - \theta)^2}{2\sigma^2}\right\} \quad \exp\left\{-\frac{\left(\frac{\sum y_i}{n} - \theta\right)^2}{2\sigma^2/n}\right\}$$

$$p(\theta) \text{ is } N(\underline{\mu}_0, \tau_0^2) \quad p(\theta \mid y_1, \dots, y_n) \sim N(\mu_1, \tau_1^2)$$

$\bar{y}$  is an estimator of  $\theta$ .

**Take**  $n = 1$ :

$$\mu_1 = \frac{\frac{\mu_0}{\tau_0^2} + \frac{y}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{1}{\sigma^2}} \quad \boxed{\bar{y} = y} \quad \text{Precision} = \frac{1}{\text{Variance}}$$

- $\boxed{\mu_1 \equiv \mu_0}$  if  $y = \mu_0$  or  $\tau_0^2 = 0$  (&  $\sigma^2 \neq 0$ )
- $\mu_1 = y$  if  $y = \mu_0$  or  $\sigma^2 = 0$ ,  $\tau^2 \neq 0$
- $\mu_1$  is the weighted average of  $\mu_0$  &  $y$ .
- Let's take  $\mu_0 = 0$ ,  $\mu_1 = \frac{y \tau_0^2}{\sigma^2 + \tau_0^2}$

$$\mu_1 = \frac{y \tau_0^2 / n}{\sigma^2 / n + \tau_0^2}$$

$$\tau_0^2(n) = \frac{\sigma^2}{n}$$

$$p(\theta \mid y) \propto p(\theta) \prod_{i=1}^n p(y_i \mid \theta)$$

## Predictive Distribution

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta) p(\theta | y) d\theta$$

$$\propto \int \exp\left(-\frac{1}{2\sigma^2}(\tilde{y} - \theta)^2\right) \exp\left(-\frac{1}{2\tau_1^2}(\theta - \mu_1)^2\right) d\theta$$

pdf of bivariate normal  $(\tilde{y}, \theta)$ .

**Q:**  $p(y) = ?$  (Marginal of  $y$ )

$$\theta \sim N(\mu_0, \tau_0^2), \quad y \sim N(\theta, \sigma^2) \quad \text{then what is } p(y)?$$

## Lecture 6

### Recap and the Marginal Distribution of $y$

$$(\cdot \dots X_n) \quad \theta \mid x_1 \dots x_n \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\frac{1}{\tau_0^2} \mu_0 + \frac{n}{\sigma^2} \bar{y}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}$$

$$p(y, \underline{\theta}) \propto \exp \left\{ - \left( \begin{pmatrix} \theta \\ y \end{pmatrix} - \mu \right)^T \Sigma^{-1} \left( \begin{pmatrix} \theta \\ y \end{pmatrix} - \mu \right) \right\}$$

$$\theta \sim N(\mu_0, \tau_0^2), \quad y \mid \theta \sim N(\theta, \sigma^2)$$

$$\text{then } p(y) = \int p(y \mid \theta) p(\theta) d\theta = \underbrace{\int \frac{\exp(-\frac{1}{2\sigma^2}(y-\theta)^2)}{\sigma\sqrt{2\pi}} \cdot \frac{\exp(-\frac{1}{2\tau_0^2}(\theta-\mu_0)^2)}{\tau_0\sqrt{2\pi}} d\theta}_{\text{Marginal Distribution of } y}$$

$$y \mid \theta \sim N(\theta, \sigma^2)$$

$$y \mid \theta = \theta + N(0, \sigma^2)$$

$$y = \theta + \varepsilon \quad = \int p(y, \theta) d\theta$$

Marginal distribution of  $\theta \sim N(\mu_0, \tau_0^2)$ ,  $y$  is Normal distribution

$$y \sim N(\mu_0, \tau_0^2 + \sigma^2)$$

$y = \theta + \varepsilon$  where  $\theta \sim N(\mu_0, \tau_0^2)$ .

$\begin{pmatrix} \theta \\ \varepsilon \end{pmatrix}$  is a bivariate normal r.v.  $\varepsilon \sim N(0, \sigma^2)$  &  $\theta, \varepsilon$  are independent.

### The Joint (Multivariate Normal) Formulation

Let  $y_1, \dots, y_n \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$  &  $\theta \sim N(\mu_0, \tau_0^2)$ . What is  $p(y_1, \dots, y_n)$ ?

$$y_1 = \theta + \varepsilon_1, \quad \dots, \quad y_n = \theta + \varepsilon_n$$

$$\left. \varepsilon_1, \dots, \varepsilon_n \text{ are iid } N(0, \sigma^2) \right\} \text{ independent of } \theta \sim N(\mu_0, \tau_0^2)$$

$$\begin{pmatrix} \theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix} \sim MVN(\mathcal{M}, \Sigma), \quad \mathcal{M} = \begin{pmatrix} \mu_0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \tau_0^2 & 0 & & & \\ 0 & \sigma^2 & 0 & & \\ & 0 & \sigma^2 & 0 & \\ & & 0 & \sigma^2 & 0 \\ & & & & \ddots \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = A \begin{pmatrix} \theta \\ \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 & 0 & 0 & \dots \\ 1 & 0 & 1 & 0 & \dots \\ 1 & 0 & 0 & 1 & \dots \\ \vdots & & & & \ddots \end{pmatrix}, \quad y = Ax$$

$$y = AX \implies y \sim MN(AM, A\Sigma A^T)$$

$$\text{Cov}(\theta + \varepsilon_1, \theta + \varepsilon_2) = \tau^2$$

$$\theta \mid y_1, \dots, y_n \sim N(\mu_1, \tau_1^2)$$

$$y \mid \theta \sim N(\theta, \sigma^2) \quad \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim MN \left( \begin{pmatrix} \mu_0 \\ \vdots \\ \mu_0 \end{pmatrix}, \begin{pmatrix} \tau^2 + \sigma^2 & \tau^2 & \cdots \\ \tau^2 & \tau^2 + \sigma^2 & \\ \vdots & & \ddots \end{pmatrix} \right)$$

$$y \mid \theta = \underline{Z} + \underline{\theta}, \quad Z \sim N(0, \sigma^2)$$

### Prediction for Unobserved Data

$$\theta \sim N(\mu_0, \tau_0^2)$$

$$y_1, \dots, y_M \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$$

Given  $y_1, \dots, y_n$ , with  $y_M$  unobserved:

$$p(y_M \mid y_1, \dots, y_n) = \int \underline{p(y_M \mid \theta)} \underline{p(\theta \mid y_1, \dots, y_n)} d\theta \quad y_M \mid y_1, \dots, y_n \sim N(\mu_1, \sigma^2 + \tau_1^2)$$

$$\int p(y \mid \theta) \underline{p(\theta)} d\theta \quad y_{n+1}, \dots, y_M \mid \underbrace{y_1, \dots, y_n}_{\text{observed}} \longrightarrow \sum_{k=1}^{M-n} y_{n+k}$$

$$\tau_1^2 = \frac{1}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \longrightarrow 0 \quad \left( \text{so } p(y_1, \dots, y_n) \longrightarrow \delta_{\theta_0} \text{ as } n \rightarrow \infty \right)$$

Handwritten notes:

$$y_{n+1}, \dots, y_M \mid (y_1, \dots, y_n) \rightarrow \bar{y}_i$$

$$\begin{pmatrix} \tau_1^2 + \sigma_1^2 & \tau_1^2 & & \\ \tau_1^2 & \tau_1^2 + \sigma_1^2 & & \\ \vdots & & \ddots & \\ & & & \tau_1^2 + \sigma_1^2 \end{pmatrix}$$

$$\sum_{k=1}^{M-n} y_{n+k}$$

## The Beta–Bernoulli Model Revisited (General $\alpha, \beta$ )

$$X_1, \dots, X_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Ber}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta)$$

$$P(X_1 = 1) = \frac{\alpha}{\alpha + \beta}$$

$$P(X_1, \dots, X_n) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{\sum X_i + \alpha - 1} (1 - \theta)^{n - \sum X_i + \beta - 1} d\theta = \frac{B(\alpha + \sum X_i, \beta + n - \sum X_i)}{B(\alpha, \beta)}$$

## A Note on Independence

$$P(X_1 = 1, X_2 = 1) \neq P(X_1 = 1) P(X_2 = 1)$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

## Sampling algorithms (mentioned in passing):

- Metropolis–Hastings algorithm
- MCMC (Markov Chain Monte Carlo)
- HMC (Hamiltonian Monte Carlo)

## Exponential Family of Distributions

$$p(y_i \mid \theta) = f(y_i) g(\theta) \exp \{ \phi(\theta)^T u(y_i) \} \quad \text{e.g. Poisson, Normal, Binomial, Exponential}$$

$$p(y_1, \dots, y_n \mid \theta) = \prod_{i=1}^n f(y_i) (g(\theta))^n \exp \left\{ \phi(\theta)^T \sum_{i=1}^n u(y_i) \right\}$$

Let  $y = y_1, \dots, y_n$ . Suppose the prior is chosen of the conjugate form

$$p(\theta) \propto (g(\theta))^\nu \exp \{ \phi(\theta)^T v \}$$

Then,

$$P(A) \propto P(B) \exp(P(A) \bar{v})$$

$$p(\theta \mid y) \propto p(\theta) \left[ (g(\theta))^n \exp \left\{ \phi(\theta)^T \sum u(y_i) \right\} \right]$$

$$\text{Then, } p(\theta \mid y) \propto (g(\theta))^{\nu+n} \exp \left\{ \phi(\theta)^T \left( \sum u(y_i) + v \right) \right\}$$

Show that the Binomial is an exponential family of distribution. Hence put an appropriate conjugate prior over it.

$$\text{Beta}(\alpha, \beta) \quad \boxed{\frac{\alpha}{\alpha + \beta}}$$

$$[\text{Natural parameter is } \log(\theta/(1 - \theta))]$$